

IDENTIFICATION OF TRANSCRIPTION FACTOR CO-BINDING PATTERNS WITH NON-NEGATIVE MATRIX FACTORIZATION

Ieva Rauluseviciute¹
Sarvesh Nikumbh^{4,5}

Timothée Launay¹
Boris Lenhard^{4,5}

Guido Barzaghi^{2,3}
Arnaud Regis Krebs²

Jaime A. Castro-Mondragon¹

Anthony Mathelier^{1,6,7,*}

¹ Centre for Molecular Medicine Norway (NCMM), Nordic EMBL Partnership, University of Oslo, 0318 Oslo, Norway

² European Molecular Biology Laboratory (EMBL), Genome Biology Unit, Meyerhofstraße 1, 69117 Heidelberg, Germany

³ Collaboration for Joint Ph.D. degree between EMBL and Heidelberg University, Heidelberg, Germany

⁴ MRC London Institute of Medical Sciences, Du Cane Road, London, W12 0NN, UK

⁵ Institute of Clinical Sciences, Faculty of Medicine, Imperial College London, Hammersmith Hospital Campus, Du Cane Road, London, W12 0NN, UK

⁶ Department of Medical Genetics, Institute of Clinical Medicine, University of Oslo and Oslo University Hospital, Oslo, Norway

⁷ Center for Bioinformatics, Department of Informatics, Faculty of Mathematics and Natural Sciences, University of Oslo, Oslo, Norway

* Correspondence: Anthony Mathelier <anthony.mathelier@ncmm.uio.no>

ADDITIONAL FILE 2

MATERIALS & METHODS

We downloaded intron coordinates for the human genome (*hg38* assembly) using the UCSC Table Browser (1). We then extracted the first two nucleotides of the exons flanking the introns. Depending on the direction of transcription, we defined these regions as donors or acceptors. We created two BED-formatted files - one for the donor and one for the acceptor sites, which we used as independent input files for COBIND. The pipeline was applied using default parameters.

RESULTS

We ran COBIND on human donor and acceptor sites to see if COBIND discovers the expected splicing motifs (2). The acceptor motif consensus in different species is GT, and the consensus motif for donor sites is CAG (2, 3).

We anchored the COBIND analyses at the two exonic nucleotides flanking the introns. Thus, we analyzed the flanking regions upstream and downstream from the exon-intron borders. One flank is in the intron and

another is in the exon (Materials and Methods above). COBIND revealed two patterns in the donor sites, where both have the consensus donor GT motif in around 7% of the input donor regions (Figure S40A). This consensus motif downstream of the anchor indicates the intronic flank. Moreover, the recovered anchor motif for the GTA motif (Co-binding pattern 0 in Figure S40A) contained high IC for AG, meaning that 4.4% of all input donor sites include AG as the last two exon nucleotides. When analyzing the acceptor sites, we found one DNA pattern corresponding to the consensus acceptor motif CAG in 1.4 % of input sequences (Figure S40B).

These analyses indicate that COBIND recovers the expected splicing signal motifs.

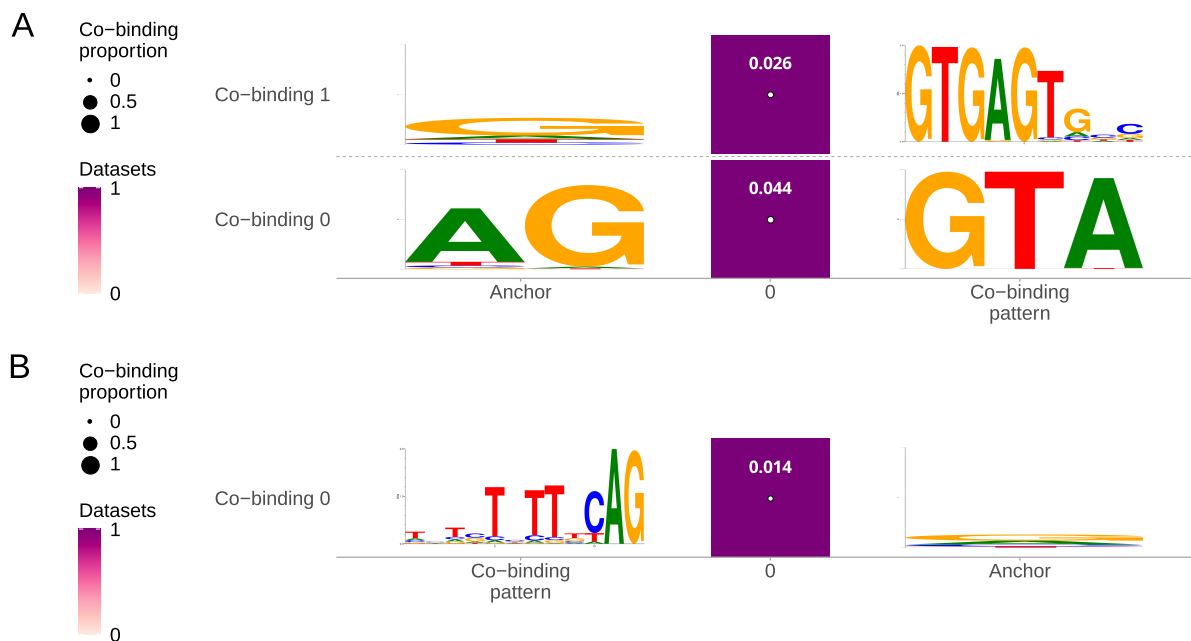


Figure S40. COBIND recovers splicing signal motifs at the intron-exon boundaries for donor (A) and acceptor (B) sites.

REFERENCES

1. Karolchik,D., Hinrichs,A.S., Furey,T.S., Roskin,K.M., Sugnet,C.W., Haussler,D. and Kent,W.J. (2004) The UCSC table browser data retrieval tool. *Nucleic acids research*, **32**, D493–D496.
2. Li,Y., Xu,Y., Ma,Z., *et al.* (2017) Comparative analysis of the exon-intron structure in eukaryotic genomes. *Yangtze Medicine*, **1**, 50.
3. Mishra,A., Siwach,P., Misra,P., Dhiman,S., Pandey,A.K., Srivastava,P. and Jayaram,B. (2021) Intron exon boundary junctions in human genome have in-built unique structural and energetic signals. *Nucleic acids research*, **49**, 2674–2683.